Al Muqshith Shifan

Applied ML / RL Developer • Systems Engineering • Research Assistant

+1 647-540-4866 mohamealmusqi.mohamedshifan@ontariotechu.net linkedin.com/in/al-mohamed-shifan github.com/Al-Scripting

Professional Summary

MSc Computer Science candidate specializing in **software design for artificial intelligence and reinforcement learning (RL)** with hands-on experience building DRL training infrastructure (multi-head PPO critics, custom PyTorch environments, persona-based reward systems) and programmable-network research systems on **Intel Tofino P4** silicon. CCNA-certified; comfortable in low-level Linux/Unix stacks. Looking to contribute to GPU-native ML tooling.

Education

Ontario Tech University — Oshawa, ON
MSc in Computer Science – Software Design (AI/RL Focus) — *May 2026 – Present*

Ontario Tech University — Oshawa, ON
Bachelor of IT – Networking & Cybersecurity (Dean's List 2026) — *Graduated Jun 2026*

Publications

K. C. Chua, **A. M. M. Shifan**, A. Neshati, L. Zaman, and C. Politowski, "RIDGE: State-Conditioned Reward Blending for Behavioral Coverage in Deep RL Game Agents," *IEEE Conference on Games (CoG)*, Madrid, Spain, Sept. 2026. **Accepted.**

Work Experience

Graduate Researcher – SEGAL Lab — May 2026 – Present
Ontario Tech University — *Oshawa, ON*

- Research on deep reinforcement learning for game agents and emotion-grounded memory for LLM-driven NPCs; co-authored work accepted at **IEEE CoG 2026**.
- Own the full research stack end to end: PyTorch training infrastructure, reproducible experiment sweeps, evaluation harnesses with cross-validated tuning and permutation-based significance testing, and the papers themselves.

Graduate Researcher – Code&SorceryLab — 2025 – Present
Independent research collective — *Oshawa, ON*

- Maintain the group's public research codebase — RIDGE, PEAK, RANGE, Procedural-Graph-Tools and the Steam patch-notes dataset — with reproducible training pipelines, pinned configs and documented experiment protocols.
- Set the engineering standards the research runs on: deterministic seeding, versioned experiment sweeps, and evaluation harnesses written before results are claimed.

Teaching Assistant – Networking I & Advanced Networking II — Sep 2025 – Present
Ontario Tech University — *Oshawa, ON*

- Led weekly lab sessions for ∼120 undergraduates across two networking courses, guiding hands-on Cisco IOS configuration of routers, switches, and stateful firewalls; mentored students through TCP/IP, VLSM subnetting, and routing protocol debugging in office hours.

Network Administrator – Summer Research Assistant — Jun 2025 – Sep 2025
Ontario Tech University — *Oshawa, ON*

- Engineered programmable-network research infrastructure on **Intel Tofino silicon P4 switches**, supporting faculty research on in-network telemetry and traffic analysis.
- Authored custom P4 data-plane programs extracting per-flow header and timing telemetry at line rate, delivering the lab's first working Tofino-deployed implementation.
- Diagnosed and patched recurring Linux kernel and driver issues across P4 control-plane servers, reducing reported system crashes by ∼30% over a 3-month window.

Cyber Security Analyst — Jun 2025 – Sep 2025
YYC Beeswax Ltd — *Remote*

- Performed vulnerability assessments on a WooCommerce + Stripe e-commerce stack, identifying 15 exploitable issues across customer-facing endpoints with prioritized remediation steps.
- Audited 30+ WordPress/WooCommerce plugins against CVE databases and authored a hardening checklist adopted as the engineering team's pre-deploy review gate.

Networking Lab Technician — Sep 2024 – Jun 2025
Ontario Tech University — *Oshawa, ON*

- Cut lab network downtime by ∼20% for 50+ daily users by isolating switching bottlenecks and reconfiguring VLAN trunking; maintained 20+ workstations and DHCP/DNS/NTP services, resolving Tier-2 tickets.

Projects

RIDGE – Reactive Inter-persona Dynamic Goal Engine | *Python, PyTorch, Crafter, PPO* 2026

- Second author on a five-author deep RL architecture for state-conditioned reward blending across four persona objectives (Explorer, Survivor, Craftsman, Warrior) on Crafter. **Accepted at IEEE Conference on Games (CoG) 2026**, Madrid, Spain. GitHub: Code-SorceryLab/RIDGE.
- Designed a **multi-head PPO critic** (shared CNN encoder, four persona-specific value heads) trained jointly with smooth sigmoid weight blending over a 6-dim game-state vector, normalized to a simplex, to address PPO non-stationarity under dynamic reward weighting.
- Built full training pipeline in PyTorch with TensorBoard logging, YAML configs, and a 4-condition × 5-seed × 1M-step experimental sweep against fixed-persona baselines.

Emotional Memory in LLM-Driven NPCs | *Systematic mapping study; in preparation, IEEE ToG* 2026

- Sole-reviewer systematic mapping study of emotional memory in LLM-driven game characters: screened 66 records against pre-registered inclusion criteria, coded the 30 included studies across six dimensions, and executed formally verified search strings against Scopus, IEEE Xplore, and the ACM Digital Library.
- Central finding is a **regression, not a gap**: pre-LLM architectures (FLAME, FAtiMA, Prom Week) coupled emotion to memory, but the field-standard memory stream dropped the emotion term and only 3 of 30 systems restored it.

EMBR – Emotional Memory for Believable Roleplay | *Python, SQLite, BM25 + embeddings* 2026

- Middleware giving NPCs persistent, emotion-grounded memory: hybrid BM25 and embedding retrieval, an affect-appraisal layer over a valence–arousal circumplex, and SQLite persistence behind a dependency-light core.
- Built the evaluation harness alongside it — leave-one-query-out cross-validated weight tuning applied identically to every baseline, exact permutation tests with Holm-Bonferroni correction, a 20-probe adversarial memory-injection corpus, and 115 passing tests.

TAST – Trait Activation Steering | *Python, PyTorch, LLM activation engineering* 2025–2026

- Per-trait activation steering for looped LLM-driven characters: steering personality expression through internal activations rather than prompt text, so persona holds across long conversations instead of drifting.
- Under revision for resubmission following AIIDE 2026 review; currently adding a prompting baseline and an audit pass over the steering-hook measurement pipeline.

PEAK – Platformer Engine for RL Benchmarking | *Python, Gymnasium, Stable-Baselines3, Hydra* 2025–2026

- Co-developed a deterministic high-performance DRL benchmarking engine with modular persona-based reward shaping, evaluating PPO agents across 11 custom 2D platformer stages with reproducible SMB1-style physics. GitHub: Code-SorceryLab/PEAK-DRL-Tool.
- Built a **Dual Spatial Hashing** collision system (separate static/dynamic grids) achieving O(C) per-query cost and sustained **1000+ environment steps per second**, making 10M-step training runs practical on single-machine hardware.

Gestura – Gesture-Based Medical Screening | *Python, Flask, CV, Gemini API, Three.js* Jan 2026

- Built at **HackHive 2026** a non-verbal patient intake tool: Flask backend streaming real-time hand-tracking state to a Three.js 3D "digital twin" frontend, with **Google Gemini** converting gestures into SOAP-style clinician summaries.

Technical Skills

Languages: Python, C++, SQL, Bash, JavaScript (Node.js/Three.js), HTML/CSS, P4
ML / RL: PyTorch, Stable-Baselines3, Gymnasium, PPO, Crafter, TensorBoard, Hydra, NumPy, Pandas, Scikit-Learn
Research Methods: Experimental design, cross-validated tuning, permutation testing, bootstrap CIs, ablation studies, systematic mapping studies
AI Systems & Web: Gemini API, Ollama, LLM prompt engineering, Computer Vision, Flask, REST APIs, Docker, Git
Networking & Sec: Intel Tofino P4, Wireshark, Nmap, Cisco IOS, Linux (Kali/RedHat), Vulnerability Assessment

Certifications: Cisco Certified Network Associate (CCNA) 2024–2027 • TryHackMe 200+ Hours (2023–Present) • Dean's List 2024-2026